

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

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if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(GP_FFCutsFits);
for i = 2:numel(fn)
    selectCuts = fn{i};
    if contains(selectCuts, 'msk')
        continue
    end
    if contains(selectCuts, 'all_fit_par')
        continue
    end

    fprintf('Processing cuts: %s \n', selectCuts);
    %selectCuts= 'Ei400_off200GPmaxFF';
    cuts_list = GP_FFCutsFits.(selectCuts);
    if ~iscell(cuts_list)
        cuts_list = {cuts_list};
    end
    if contains(selectCuts, 'Ei200')
        mi = maps_instrument(200,600, 'S');
    end
    if contains(selectCuts, 'Ei400')
        mi = maps_instrument(401,600, 'S');
    end
    if contains(selectCuts, 'Ei800')
        mi = maps_instrument(786,600, 'S');
    end
    sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
    for j=1:numel(cuts_list)
        cuts_list{j} = cuts_list{j}.set_instrument(mi);
        cuts_list{j} = cuts_list{j}.set_sample(sample);
        plot(cuts_list{j}); lz 0 1; keep_figure;
    end
    dir_name = "GP";

    dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
    half_dE = 5; % half width of data binning

    en_mm = cuts_list{1}.data.img_range(:,4);

```

```

en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
cl = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep

all_fit_par = GP_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;

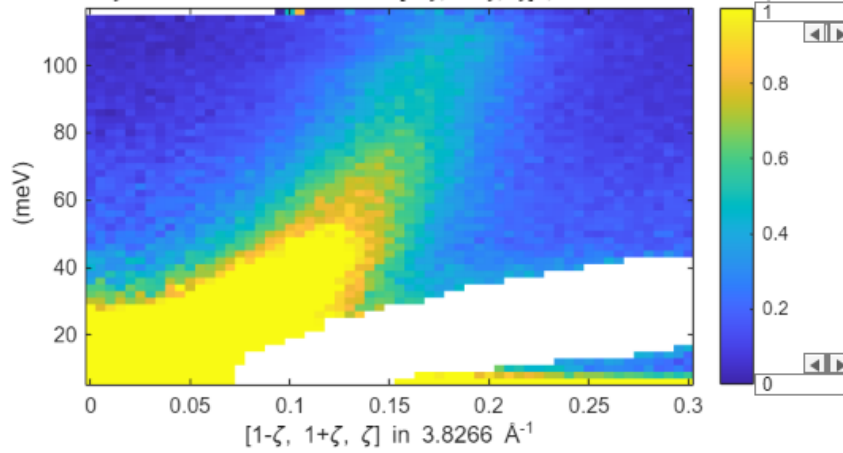
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
close all;
end

```

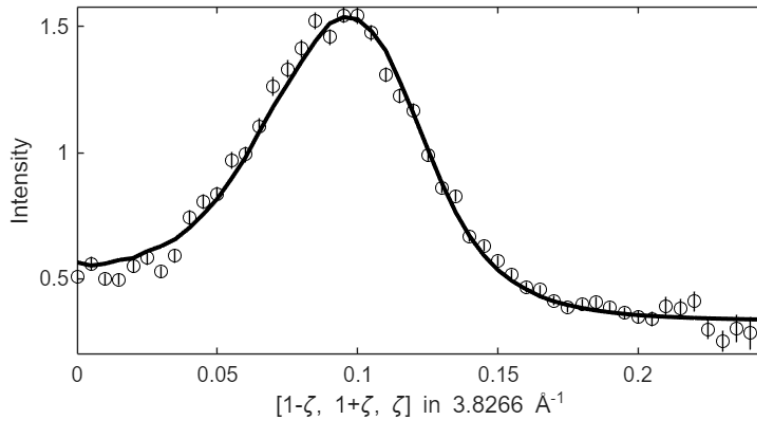
Processing cuts: Ei200_off110Gpmiff

Fe_ei200_align_sym3, w2_200off110m.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$, $E=5:2:117$

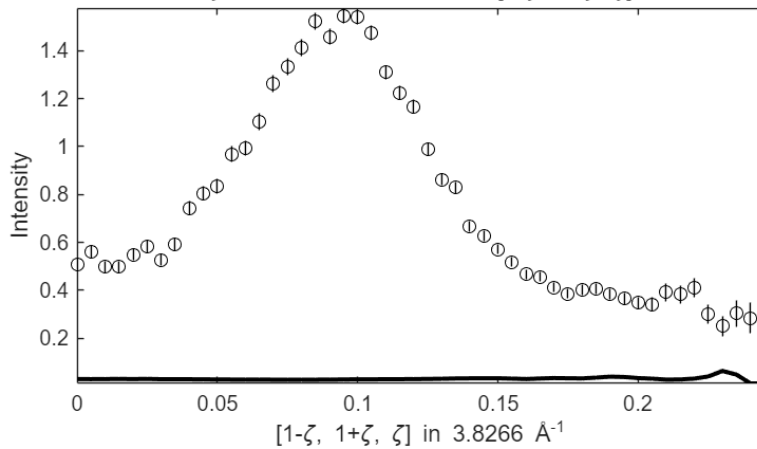


 **** En = 35 ± 5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.87 \pm 0.017$; $J0=24 \pm 0.15$; $\gamma=6.26505 \pm 0.127623$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $30 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

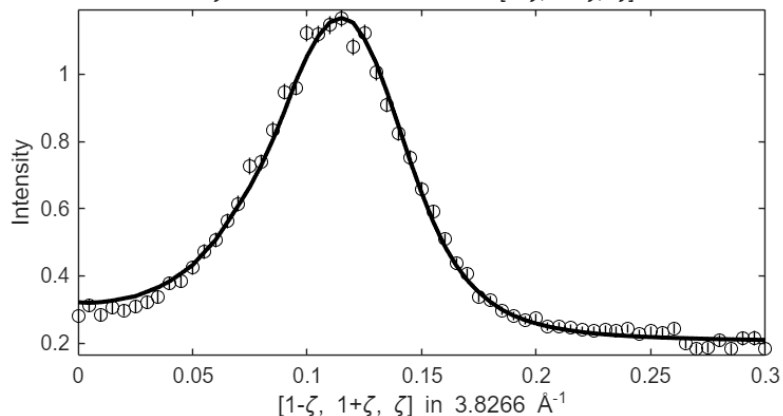


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $30 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

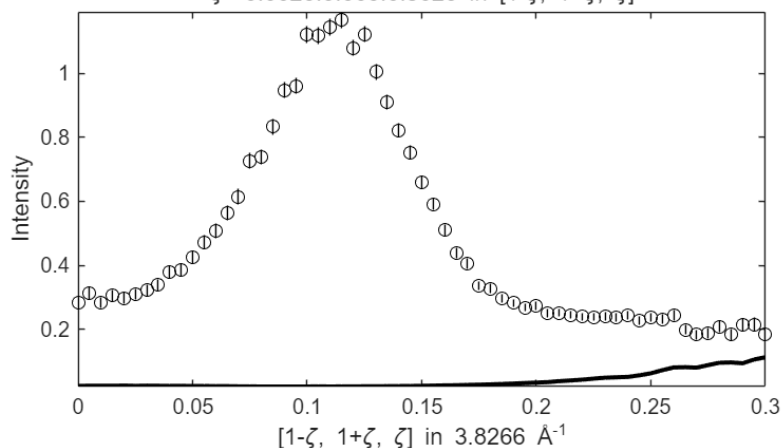


 **** En = 45 ± 5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.88\pm0.011$; $J_0=25\pm0.11$; $\gamma=10.1715\pm0.140422$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $40 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

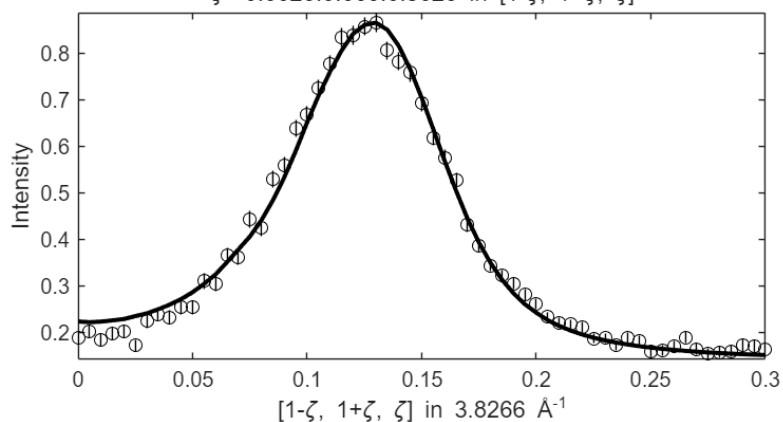


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $40 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

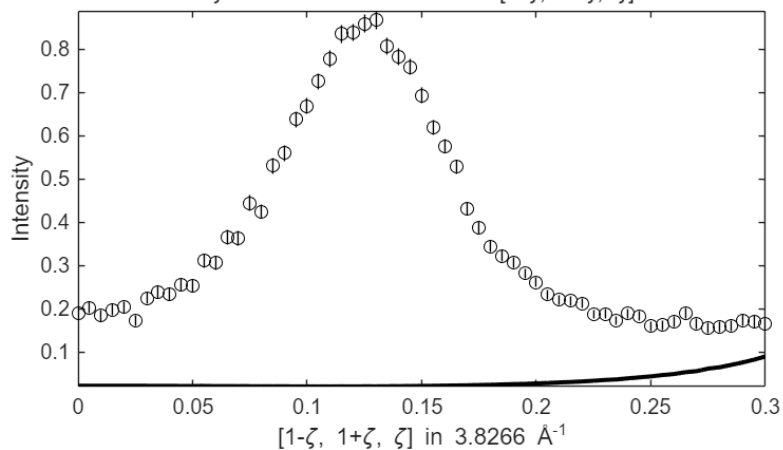


 **** En = 55±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.89\pm0.016$; $J_0=26\pm0.12$; $\gamma=16.2103\pm0.310405$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $50 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

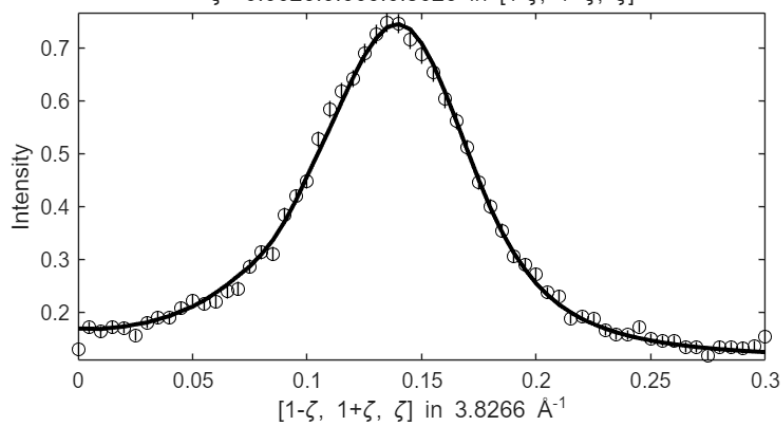


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $50 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

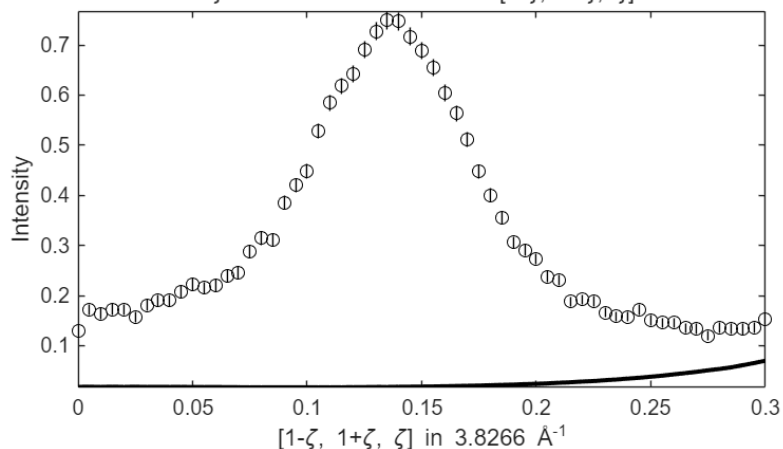


**** En = 65±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S = 0.9 \pm 0.015$; $J0 = 27 \pm 0.092$; $\gamma = 19.912 \pm 0.329451$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

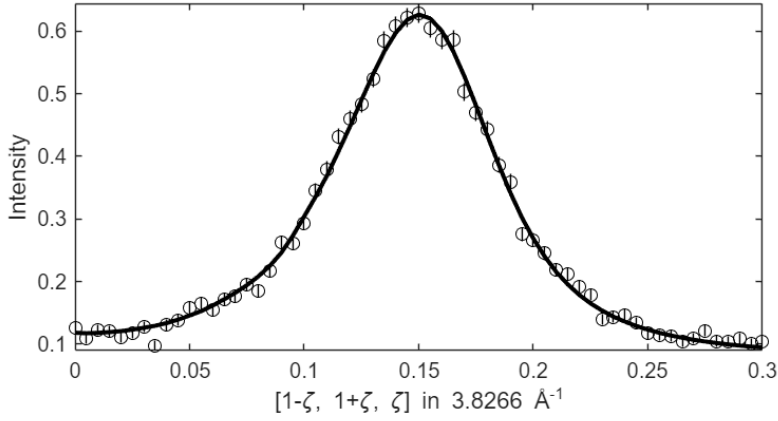


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

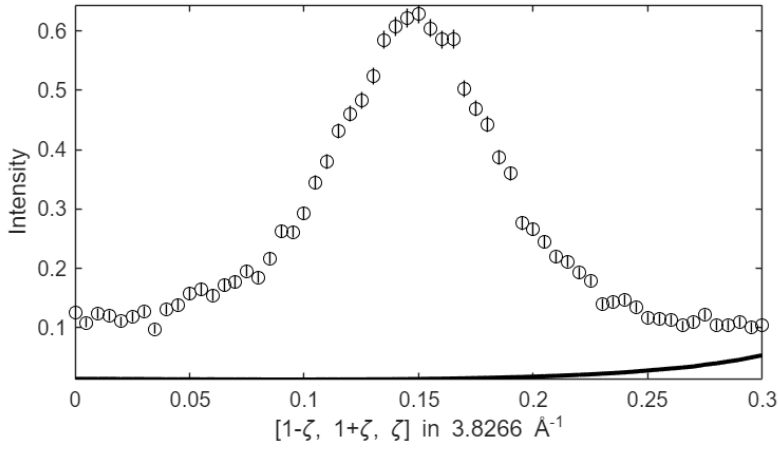


**** En = 75±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.86\pm0.018$; $J_0= 28\pm 0.1$; $\gamma=22.9969\pm0.550432$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

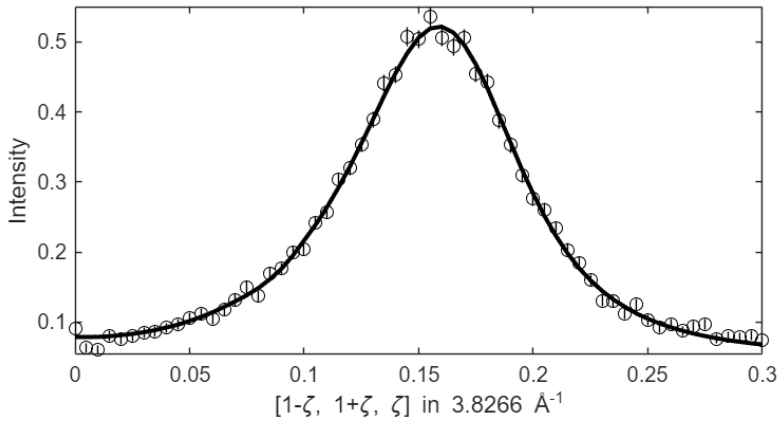


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

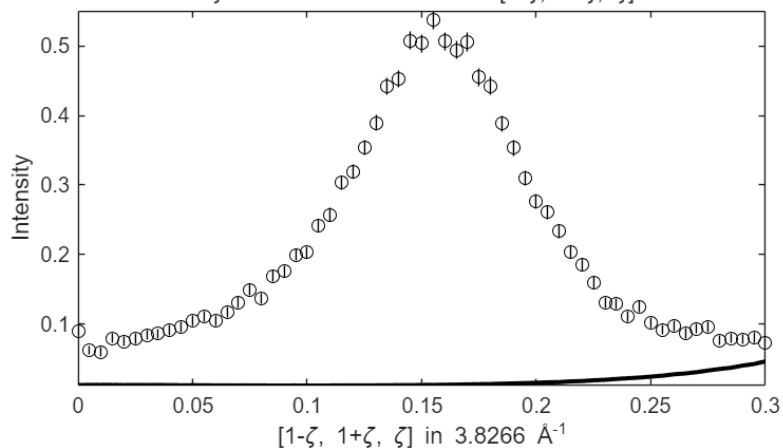


 **** En = 85±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.88\pm0.023$; $J_0= 30\pm0.13$; $\gamma=28.805\pm0.970987$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

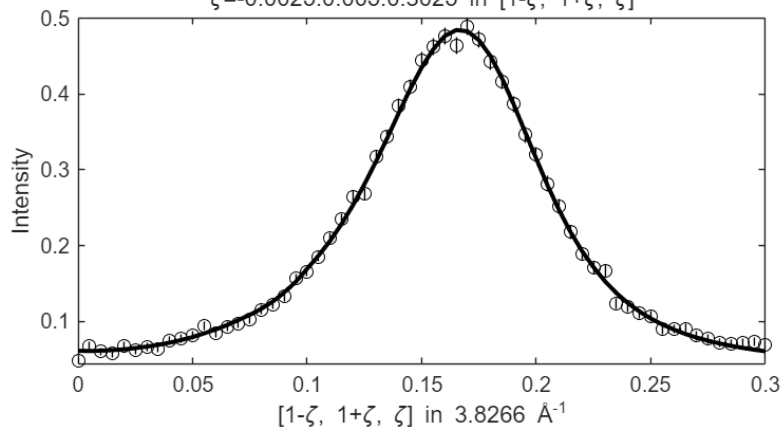


Fe_ei200_align_sym3, w2_200off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

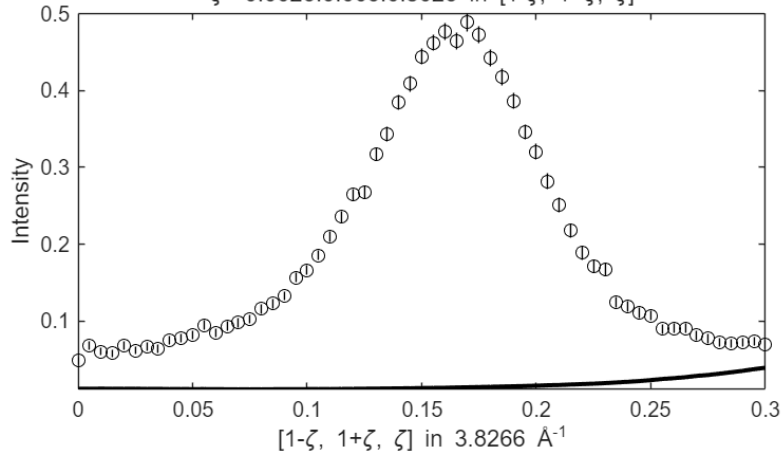


**** En = 95±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.87 \pm 0.016$; $J0=31 \pm 0.1$; $\gamma=31.5939 \pm 0.778276$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$:
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

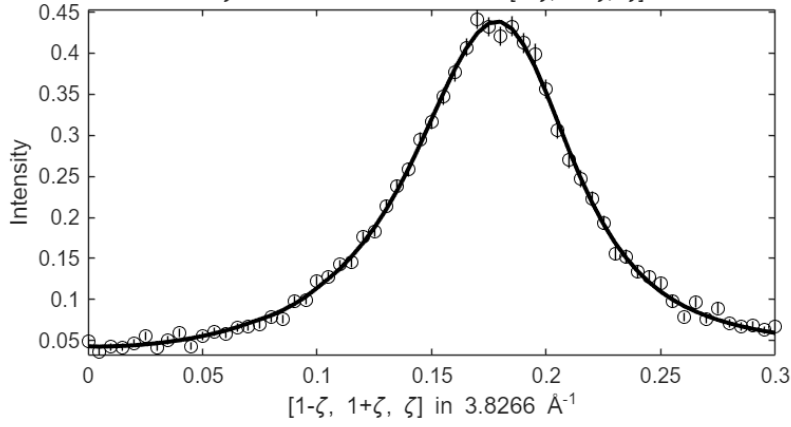


Fe_ei200_align_sym3, w2_200off110m.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$:
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

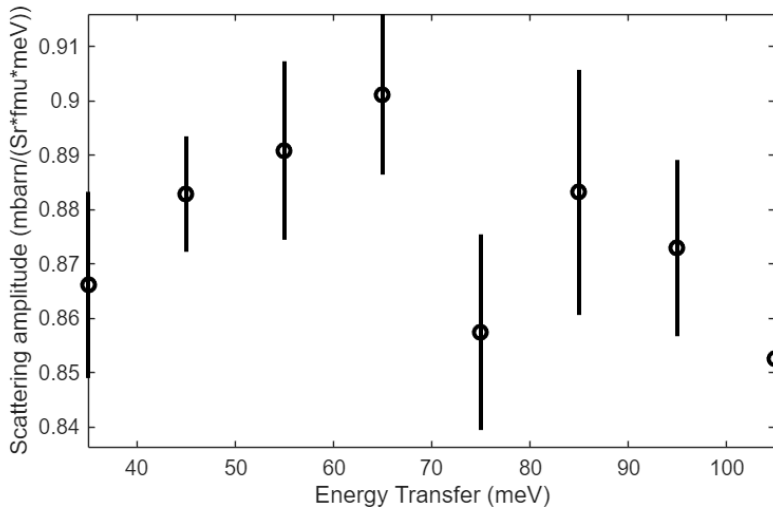
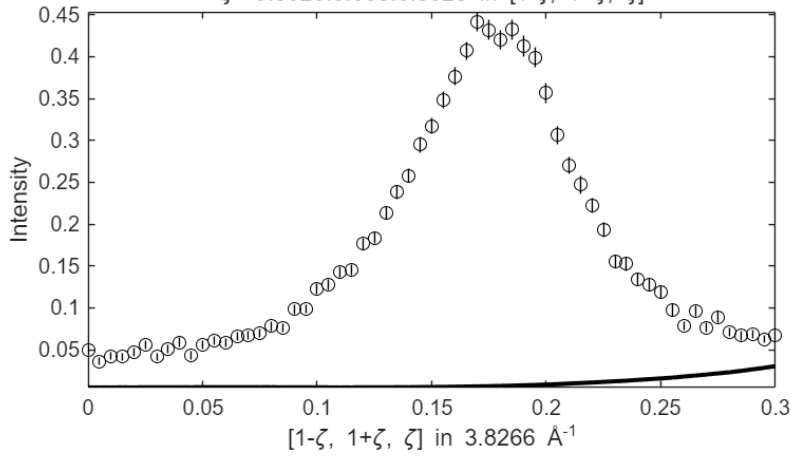


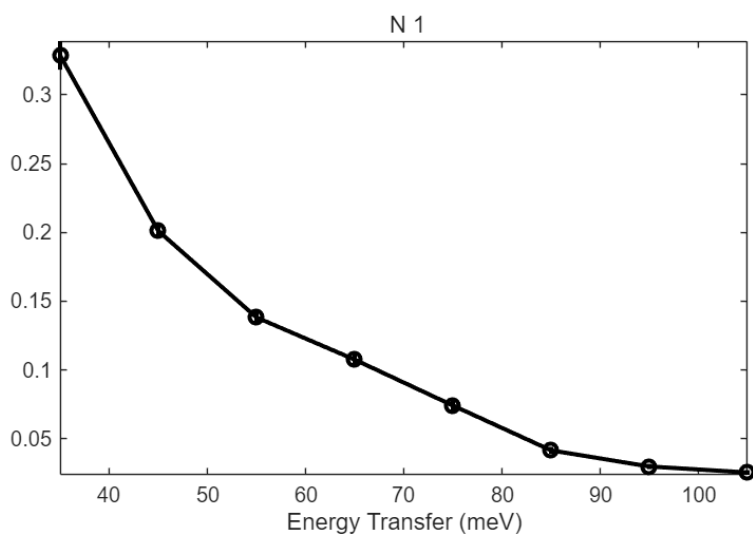
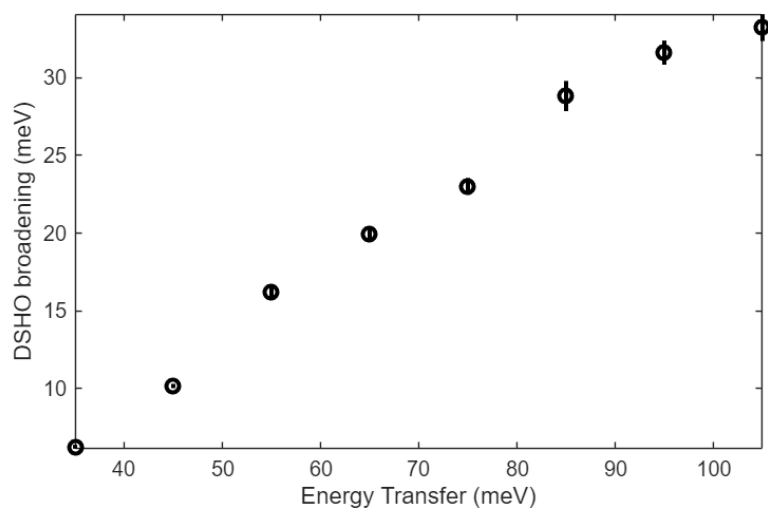
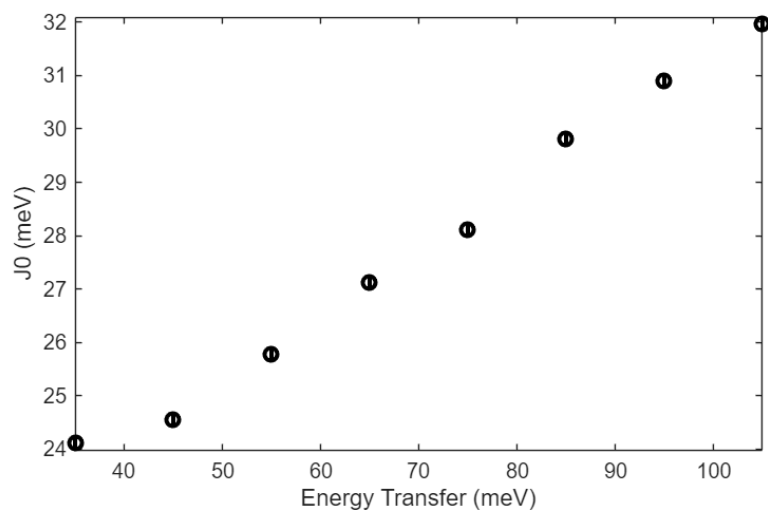
**** En = 105±5 going to 105

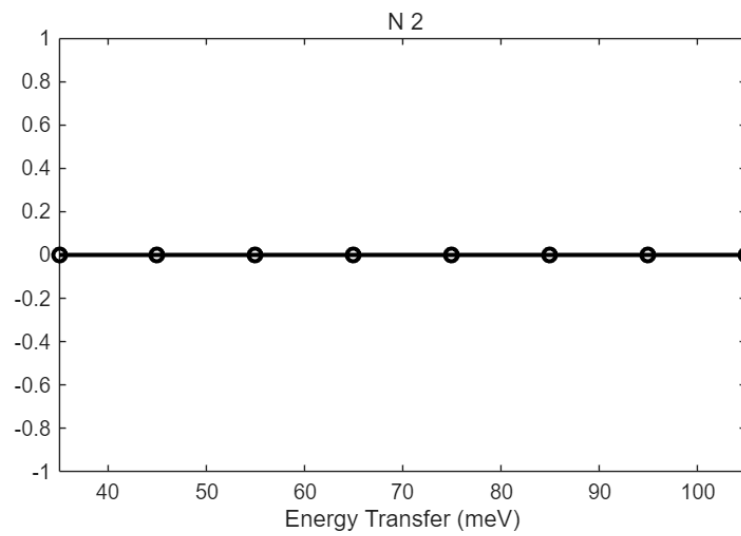
Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.85\pm0.016$; $J_0=32\pm0.12$; $\gamma=33.204\pm0.865485$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$







Processing cuts: Ei200_off200GPmiff

Index exceeds the number of array elements. Index must not exceed 1.

```
fit_src_struct = GP_FFCutsFits;  
save(Cutfitfile,"fit_src_struct",-v7.3')
```